

Lourdu Raju

Machine Learning Engineer — Production Computer Vision | GPU Inference | Applied GenAI

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PROFESSIONAL SUMMARY

ML engineer (about 2 years) who owns a utility-scale OCR system end to end, from training to TensorRT/Triton serving; lifted live accuracy **79% → 91%** over **40M** readings. Founder of SpaceDrift, a 5-person ML studio.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Bash

Modeling: PyTorch, Hugging Face, YOLO (OBB), SVTRv2, MobileViTv2, OpenCV, scikit-learn, NumPy

Inference: TensorRT (FP16), Triton Inference Server, ONNX Runtime, `torch.compile`, load testing

MLOps & Cloud: DVC, GitHub Actions, pytest, Docker, Kubernetes; AWS (EC2, Lambda, S3, DynamoDB)

GenAI: LangGraph, Ollama, RAG (Qdrant, ChromaDB, cross-encoder reranking), FastAPI, React

EXPERIENCE

Sujanix Private Limited

Jan 2026 – Present

Machine Learning Engineer

Bengaluru, India

- Own a five-model meter-reading pipeline (MobileViTv2, YOLO26n-OBB, **SVTRv2** + CTC) at up to **330K req/day**; raised live accuracy **79% → 91%** over 40M production readings.
- Built the GPU path on Triton with 9 TensorRT FP16 engines on one L4: p50 latency **1,415 → 156 ms (9×)**, **181 img/s** sustained at p95 340 ms with zero errors over 67,719 requests (12.6× peak load).
- Ported the classifier from ONNX Runtime to TensorRT after patching an unsupported UINT8 **Transpose**: compute **309.5 → 3.3 ms (94×)**, throughput 19 → 321 img/s.
- Retrained SVTRv2 on production crops (**87.7% → 90.1%** exact match on a fixed 3,965-image set) and made training **3.8× faster** with fused SDPA and static `torch.compile`.
- Root-caused a release regression to a CTC decoding bug (64.2% → 84.0% restored); added DVC-pinned releases, ONNX parity checks, benchmark-gated promotion and a canary router with serverless fallback.

SpaceDrift (spacedrift.in)

Aug 2024 – Present

Founder | MSME-registered ML & AI studio

Bengaluru, India

- Founded and lead a **5-person** studio that has delivered **10+** fixed-scope, fixed-price engagements in research ops for PhD scholars and document AI / OCR.
- Own scoping, pricing and delivery; ship reproducible, seed-controlled experiment pipelines, baseline replications and extraction pipelines with code, data and environment handed over.

BrainOvision Solutions

Feb 2024 – Apr 2024

Data Science Intern

Hyderabad, India

- Improved forecasting accuracy **15%** with ensemble gradient boosting on multi-million-row data.

PROJECTS

ECHOME — Local-First Agent Memory | *LangGraph, Qdrant, Ollama, IRT*

github.com/Lourdhu02/echome

- LangGraph agent with three-tier memory (episodic in Qdrant, LLM-consolidated semantic, procedural); adaptive IRT assessment with Fisher-information item selection cuts test length **70%** at SE < 0.32.

FinSentinelAI — Private Document RAG | *FastAPI, Ollama, ChromaDB*

github.com/Lourdhu02/fin-sentinal.ai

- Fully local RAG over invoices and bank statements with cross-encoder reranking, JWT-isolated workspaces and VLM-based ingestion; no document leaves the machine.

ACHIEVEMENTS & CERTIFICATIONS

Kaggle Notebooks Expert | ML Specialization (DeepLearning.AI, Stanford) | NPTEL Data Science (IIT Madras)

EDUCATION

Andhra Loyola Institute of Engineering and Technology

Bachelor of Technology in Computer Science and Engineering

May 2024

Vijayawada, India